BCA Digital Image Processing Lab Viva – Questions with Answers

Program 1: Load and display grayscale and colour images

1. What is the difference between a grayscale and a colour image?
A grayscale image has one intensity channel, whereas a colour image commonly has three channels.

2. Which functions are used to load and display an image?
`cv2.imread()` loads an image, and `cv2.imshow()` displays it.

3. How do you convert a colour image to grayscale?
`cv2.cvtColor(img, cv2.COLOR_BGR2GRAY).`

4. Why does OpenCV use BGR by default when reading colour images?
OpenCV conventionally stores colour channels in Blue, Green, Red order.

Program 2: Image sampling and quantization

1. What is sampling?
Sampling converts a continuous image into a discrete grid of pixels.

2. What is quantization?
Quantization maps continuous intensity values to a finite set of intensity levels.

3. What is the difference between sampling and quantization?
Sampling determines spatial resolution; quantization determines intensity resolution.

4. What happens when the number of gray levels decreases?
The image may show visible intensity bands and lose subtle details.

Program 3: Spatial and intensity resolution

1. What is spatial resolution?
It describes the spatial detail represented by the pixel grid.

2. What is intensity resolution?
It describes how many different intensity values each pixel can represent.

3. How many gray levels are possible in a 4-bit grayscale image?
2^4=16) gray levels.

4. How many gray levels are possible in an 8-bit grayscale image?
(2^8=256) gray levels.

Program 4: Adjacency and connectivity in a binary image

1. What is adjacency?
Adjacency determines whether two pixels are neighbours according to a specified neighbourhood rule.

2. What is 4-connectivity?
Pixels are connected through their horizontal and vertical neighbours.

3. What is 8-connectivity?
Pixels are connected through horizontal, vertical and diagonal neighbours.

4. What is m-adjacency?
Mixed adjacency allows diagonal connectivity when the common 4-neighbours do not create an ambiguous connection under the specified rule.

5. What is a connected component?
A group of foreground pixels connected according to a chosen adjacency rule.

Program 5: RGB components and RGB-to-CMY/HIS conversion

1. What does RGB stand for?
Red, Green and Blue.

2. What does CMY stand for?
Cyan, Magenta and Yellow.

3. How do you calculate CMY from normalized RGB values?
(C=1-R, M=1-G, Y=1-B).

4. What is the HIS colour model?
A colour representation using Hue, Intensity and Saturation. It is closely related to the more commonly named HSI model.

5. Why are colour models converted?
Different models are useful for different tasks, such as colour representation, enhancement and segmentation.

Program 6: Modify hue, saturation and intensity

1. What is hue?
Hue represents the basic colour, such as red, green or blue.

2. What is saturation?
Saturation indicates the purity or vividness of a colour.

3. What is intensity?
Intensity represents the overall brightness component in the HSI model.

4. What happens when saturation is reduced?
Colours become less vivid and move toward gray.

5. What happens when intensity is increased?
The image generally becomes brighter, subject to the representation and clipping limits.

Program 7: Colour histograms

1. What is a colour histogram?
It shows the distribution of intensity values for each colour channel.

2. How many channel histograms are usually calculated for a BGR image?
Three: blue, green and red.

3. Which OpenCV function calculates a histogram?
`cv2.calcHist()`.

4. What does a histogram tell us?
It indicates the distribution of pixel intensities but does not directly show where those pixels occur.

Program 8: Image negative transformation

1. What is image negative transformation?
A point transformation that reverses intensity values.

2. What is the formula for an 8-bit image?
\(s=255-r\).

3. What is the negative of pixel intensity 60?
\(255-60=195\).

4. What is its main application?
Highlighting details that are difficult to see in the original intensity representation.

Program 9: Log transformation

1. What is log transformation?
An intensity transformation that expands lower intensity values relative to higher ones.

2. What is its formula?
(s=clog(1+r)).

3. Why is 1 added to r?
To avoid taking the logarithm of zero.

4. What is the purpose of the constant c?
It scales the transformed intensities to a suitable range.

Program 10: Power-law or gamma transformation

1. What is gamma correction?
A nonlinear intensity transformation used to adjust brightness and tonal appearance.

2. What is its formula?

$s = cr^{\gamma}$, generally applied to normalized intensities.

3. **What happens when $\gamma < 1$?**
The image generally becomes brighter.

4. **What happens when $\gamma > 1$?**
The image generally becomes darker.

5. Why are different gamma values tested?
To compare their effects on image brightness and contrast.

Program 11-A: Contrast stretching

1. What is contrast stretching?
A transformation that expands a restricted range of intensities.

2. Why is it used?
To make details more distinguishable in low-contrast images.

3. What is the difference between contrast stretching and histogram equalization?
Contrast stretching remaps a selected range, while histogram equalization uses the cumulative histogram.

4. What happens to pixel values outside the selected range?
Depending on the implementation, they may be clipped or mapped to the output limits.

Program 11-B: Thresholding

1. What is thresholding?
Classifying pixels based on whether their intensities are above or below a threshold.

2. What is the basic thresholding formula?
s=255 if r>T; otherwise, s=0, for a typical binary threshold.

3. What does T represent?
The threshold value.

4. What is the difference between global and adaptive thresholding?
Global thresholding uses one threshold for the image; adaptive thresholding calculates thresholds for local regions.

Program 12: Histogram equalization

1. What is histogram equalization?
A contrast-enhancement method based on the cumulative distribution of pixel intensities.

2. What is CDF?
The cumulative distribution function, which accumulates histogram probabilities.

3. Which OpenCV function performs grayscale histogram equalization?
`cv2.equalizeHist()`.

4. Does histogram equalization always improve an image?
No. It can amplify noise or make some images look unnatural.

Program 13: Histogram stretching

1. What is histogram stretching?
Expanding an image's intensity range to use a wider portion of the available range.

2. What is the usual output range for an 8-bit grayscale image?
0 to 255.

3. What happens when the original intensity range is narrow?
The image often appears to have low contrast.

4. How does histogram stretching differ from histogram equalization?
Stretching remaps intensity endpoints or selected ranges; equalization redistributes intensities using the CDF.

Program 14: Histogram matching or specification

1. What is histogram matching?
Transforming an image so that its intensity distribution approximates that of a reference image.

2. What is the reference image?
The image whose histogram distribution is used as the target.

3. How does histogram matching differ from histogram equalization?
Equalization aims for a broadly spread distribution; matching aims for a specified target distribution.

4. Which Python library provides a histogram-matching function?
Scikit-image provides `skimage.exposure.match_histograms()`.

Program 15: Spatial correlation and convolution

1. What is spatial correlation?
A kernel is moved across an image and multiplied with corresponding neighbourhood values, typically without flipping the kernel.

2. What is convolution?
A similar operation in which the kernel is flipped before being applied.

3. What is a kernel?
A small matrix of coefficients used to process local image neighbourhoods.

4. Which OpenCV function performs filtering with a custom kernel?
`cv2.filter2D()`.

5. What is one practical use of convolution?
Smoothing, sharpening or edge detection.

Program 16: Mean and Gaussian smoothing filters

1. What is smoothing?
Reducing local intensity variations to suppress noise or fine detail.

2. What is a mean filter?
A filter that replaces a pixel with the average of values in its neighbourhood.

3. What is a Gaussian filter?
A weighted smoothing filter whose coefficients follow a Gaussian distribution.

4. Which OpenCV functions implement these filters?
`cv2.blur()` for averaging and `cv2.GaussianBlur()` for Gaussian smoothing.

5. What is the main disadvantage of smoothing?
It can blur edges and small details.

Program 17: Image sharpening using the Laplacian

1. What is the Laplacian operator?
A second-order derivative operator that measures changes in the intensity gradient.

2. Why is it used for sharpening?
It highlights rapid intensity changes that correspond to edges and fine details.

3. Which OpenCV function calculates the Laplacian?
`cv2.Laplacian()`.

4. Why is the Laplacian sensitive to noise?
Noise can produce rapid intensity changes that the derivative operator also detects.

Program 18: Unsharp masking and high-boost filtering

1. What is unsharp masking?

A technique that subtracts a blurred version of the image from the original to extract detail.

2. **What is the unsharp mask formula?**
$M = f - \bar{f}$, where f is the original and $\bar{f}$ is the blurred image.

3. **How is the sharpened image calculated?**
$g = f + kM$, where k controls sharpening strength.

4. What is high-boost filtering?
A sharpening method that retains a scaled contribution of the original image along with enhanced detail.

5. What happens if the sharpening strength is too high?
The image may develop exaggerated edges, halos or amplified noise.

Program 19: Sobel and Prewitt edge detection

1. What is an edge?
A location where image intensity changes significantly.

2. What is the Sobel operator?
A gradient operator that estimates intensity changes in horizontal and vertical directions.

3. What is the Prewitt operator?
Another gradient-based edge detector using simple convolution masks.

4. What do G_x and G_y represent?
The horizontal and vertical gradient components.

5. **How is gradient magnitude calculated?**
$\sqrt{G_x^2 + G_y^2}$.

6. What is the difference between Sobel and Prewitt?
Their kernel coefficients differ; Sobel gives additional weight to the central row or column.

Program 20: 2D Fourier transform and inverse DFT

1. What is the Fourier transform?
It represents an image in terms of its spatial-frequency components.

2. What is the 2D DFT used for?
Frequency analysis and frequency-domain filtering of images.

3. What is the inverse DFT?
It reconstructs the spatial-domain image from its frequency representation.

4. Which NumPy functions are used?
`np.fft.fft2()` and `np.fft.ifft2()`.

5. Why is `np.fft.fftshift()` used?
To move the zero-frequency component to the centre of the spectrum for easier visualization and filter design.

Program 21: Low-pass and high-pass frequency-domain filters

1. What is a low-pass filter?
A filter that retains low spatial frequencies and suppresses high frequencies.

2. What is a high-pass filter?
A filter that retains high spatial frequencies and suppresses low frequencies.

3. What is the effect of low-pass filtering?
Image smoothing or blurring.

4. What is the effect of high-pass filtering?
Emphasizing edges and fine details.

5. What is the difference between spatial and frequency-domain filtering?
Spatial filtering operates directly on pixels; frequency-domain filtering modifies the Fourier representation.

Program 22-A: DFT property — Translation

1. What is the translation property of the DFT?
Shifting an image in the spatial domain introduces a corresponding phase change in its Fourier transform.

2. Does a spatial shift change the magnitude spectrum?
Ideally, no. A spatial translation changes the phase but preserves the magnitude spectrum.

3. What is phase in a Fourier transform?
Phase represents the angular component of each complex frequency coefficient.

4. Why study the translation property?
To understand how image position relates to frequency-domain representation.

Program 22-B: DFT property — Scaling

1. What is image scaling?
Changing an image's spatial dimensions, making it larger or smaller.

2. What is the relationship between spatial and frequency domains under scaling?
Broadly, spatial expansion corresponds to compression of the frequency representation, and

spatial compression corresponds to expansion, subject to discrete sampling and scaling conventions.

3. Why may interpolation be required when scaling an image?
To estimate pixel values at new spatial locations.

4. Why demonstrate DFT scaling?
To understand how changes in spatial size affect the frequency representation.

Program 23: Morphological operations — Dilation, erosion, opening and closing

1. What is mathematical morphology?
A set-theoretic approach to processing shapes and structures in images.

2. What is a structuring element?
A small predefined shape used to determine how a morphological operation affects an image.

3. What is dilation?
An operation that generally expands foreground regions.

4. What is erosion?
An operation that generally shrinks foreground regions.

5. What is opening?
Erosion followed by dilation, commonly used to remove small foreground objects or protrusions.

6. What is closing?
Dilation followed by erosion, commonly used to fill small gaps or holes.

7. Which OpenCV functions perform these operations?
`cv2.dilate()`, `cv2.erode()`, `cv2.morphologyEx()` with `cv2.MORPH_OPEN` or `cv2.MORPH_CLOSE`.

Program 24: Hit-or-miss transformation

1. What is the hit-or-miss transformation?
A morphological operation used to detect specific patterns in binary images.

2. What does the operation search for?
A particular foreground configuration together with a specified background configuration.

3. What type of image is commonly used?
A binary image.

4. What is a structuring element's role in this operation?
It defines the foreground and background pattern that must match.

5. Give one application of hit-or-miss transformation.
Detecting particular shapes, patterns or configurations in binary images.

Program 25: Noise models and mean filtering

1. What is image noise?
Unwanted variations in pixel values that degrade image quality.

2. What is Gaussian noise?
Noise modelled using a Gaussian probability distribution.

3. What is salt-and-pepper noise?
Impulse noise that introduces randomly occurring very bright and very dark pixels.

4. What is Rayleigh noise?
Noise modelled by a Rayleigh probability distribution.

5. What is Gamma noise?
Noise whose values follow a Gamma probability distribution.

6. What is exponential noise?
Noise modelled by an exponential probability distribution.

7. What does a mean filter do?
It replaces each pixel with the average of its neighbourhood values.

8. Is a mean filter suitable for every type of noise?
No. It is useful for some random noise, but a median filter is often better for salt-and-pepper noise.

9. What is the difference between Gaussian and impulse noise?
Gaussian noise produces distributed intensity fluctuations; impulse noise produces isolated extreme values.

10. How can filter performance be evaluated?
By comparing visual quality or using measures such as mean squared error (MSE) and peak signal-to-noise ratio (PSNR), when a clean reference image is available.